

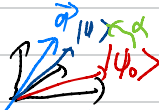
Arbitrary state preparation

Input: $|\psi_0\rangle = |0\rangle$

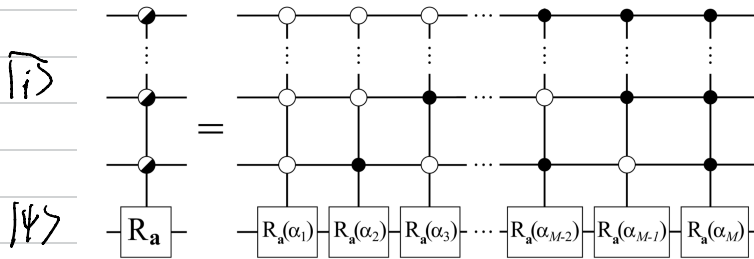
Output: $|\psi\rangle = (a_i e^{i\omega_i})$

① Single-qubit rotation $R_{\vec{a}}(\alpha)$

$$R_{\vec{a}}(\alpha) = e^{i\vec{a} \cdot \vec{\sigma} \alpha/2} = \cos \frac{\alpha}{2} + i(\vec{a} \cdot \vec{\sigma}) \sin \frac{\alpha}{2}$$



② k -controlled, m -target bit rotation $F_m^k(\vec{a}, \vec{\alpha})$



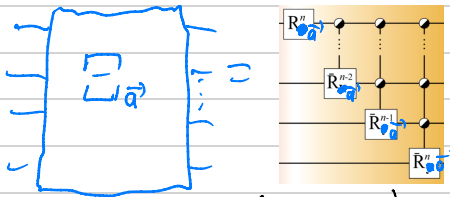
$$|\psi\rangle = \sum_{i=1}^{2^n} a_i |i\rangle \quad 2^n = 2^{n-1} \times 2$$

$$= \sum_{i=1}^{2^{n-1}} (a_{2i} |i\rangle \otimes |0\rangle + a_{2i-1} |i\rangle \otimes |1\rangle)$$

$$F_m^k(\vec{a}, \vec{\alpha}) |\psi\rangle = \sum_{i=1}^{2^{n-1}} (a_{2i} |i\rangle \otimes R_{\vec{a}}(\alpha_0) |0\rangle + a_{2i-1} |i\rangle \otimes R_{\vec{a}}(\alpha_0) |1\rangle)$$

$$F_m^0 = R_{\vec{a}}(\alpha_0)$$

③ Multi-qubit rotation $\Xi_{\vec{a}}(\{\vec{\alpha}_i\})$



$$\Xi_{\vec{a}}(\{\vec{\alpha}_i\}) = \prod_{i=0}^{n-1} F_{n-k}^{n-k-1}(\vec{a}, \vec{\alpha}_{k+1}) \otimes I_{2^k}$$

